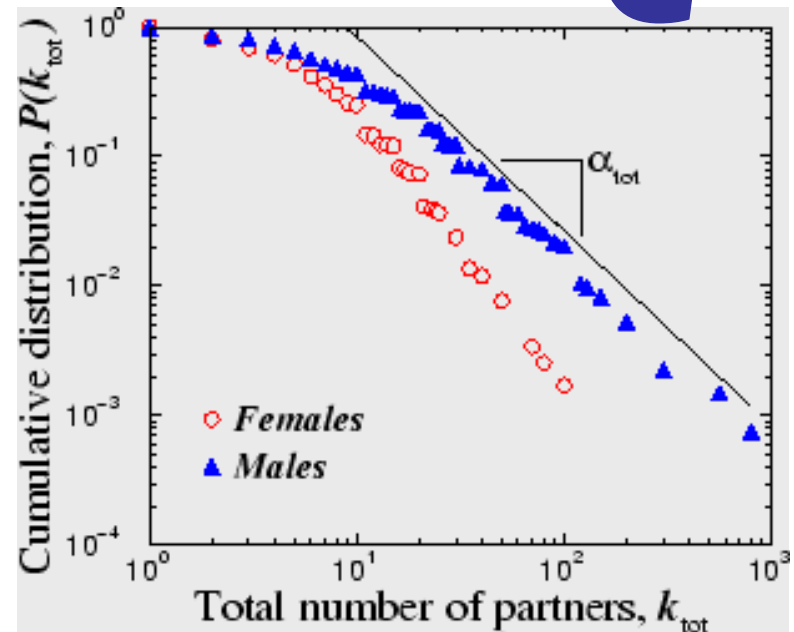


A roadmap in the world of networks

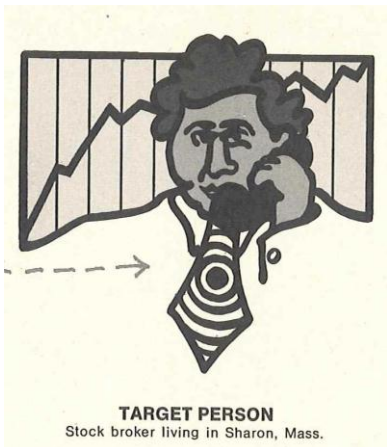
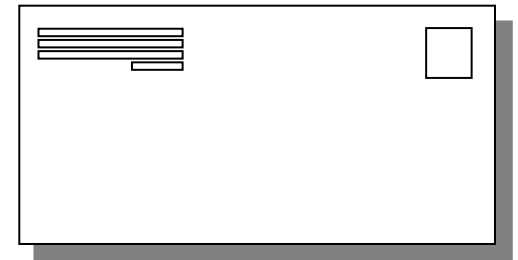
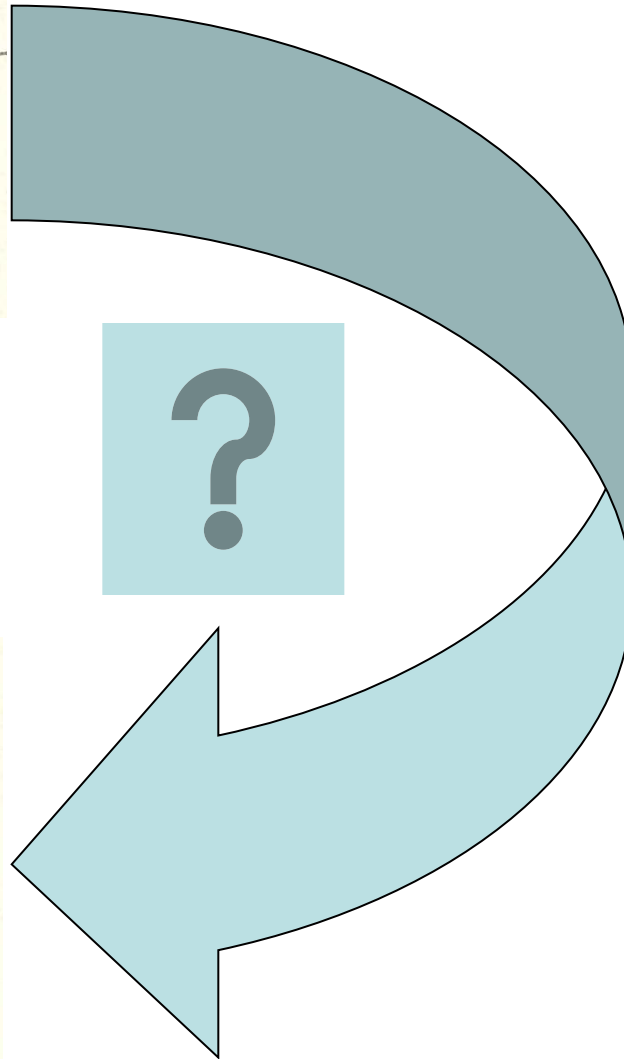
Networks and Innovation Training Programme
Dr Antoine Vernet & Dr Anne ter Wal

11 September 2013

What is the value of
studying sexual
relationships through
network analysis?



Nodes: Swedish people
Links: sexual relationships

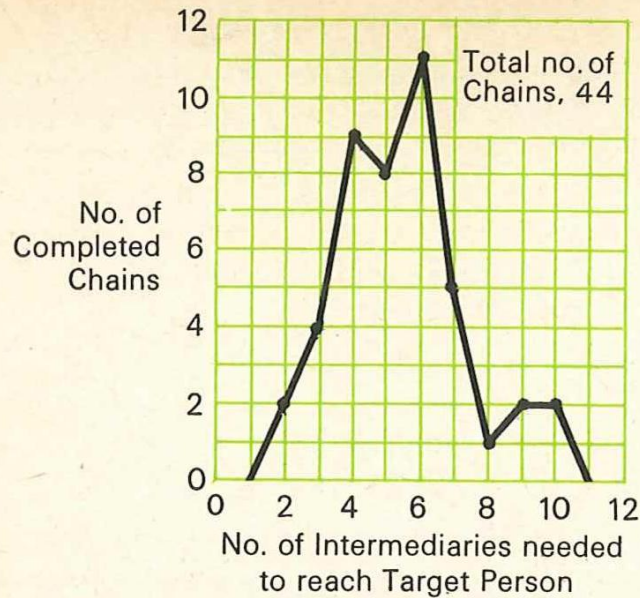
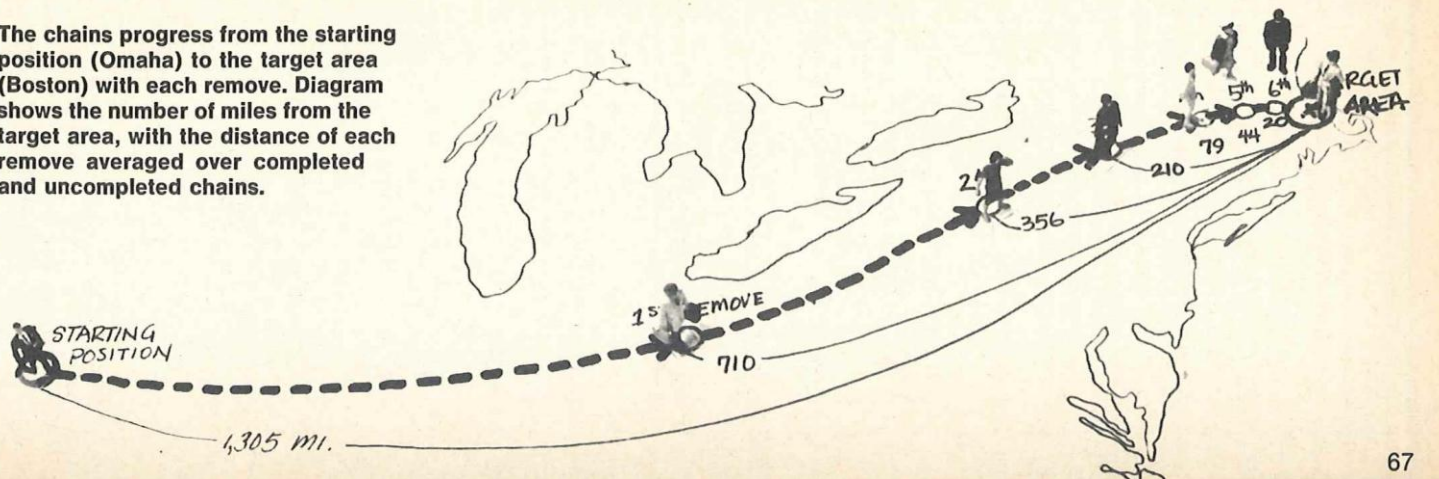


Source: Milgram (1967)



Source: Milgram (1967)

The chains progress from the starting position (Omaha) to the target area (Boston) with each remove. Diagram shows the number of miles from the target area, with the distance of each remove averaged over completed and uncompleted chains.



Source: Milgram (1967)

Degrees of separation

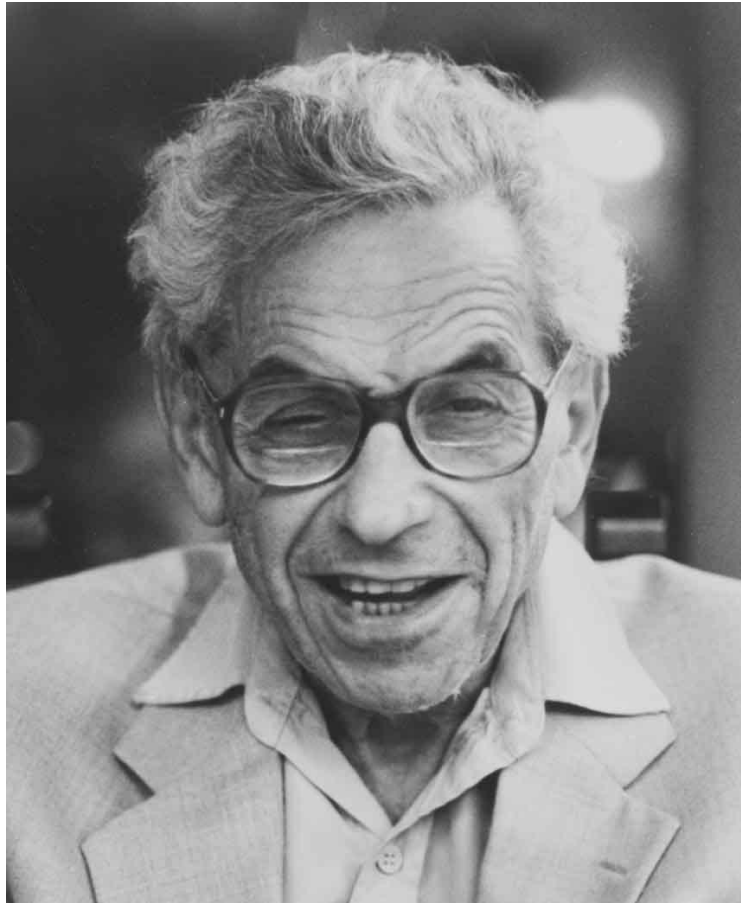


Photo by Kevin Mazur/wireimage



Degrees of separation



3.7

degrees of separation

=4.74 average PL

- Study on the **degrees of separation** on the world's largest social network (2011)
- **721 million 'active' members**, roughly 10% of the world's population
- **69 billion friendships**
average network degree: 96
- Type of tie: "**friendship**"

Degrees of separation



5.6

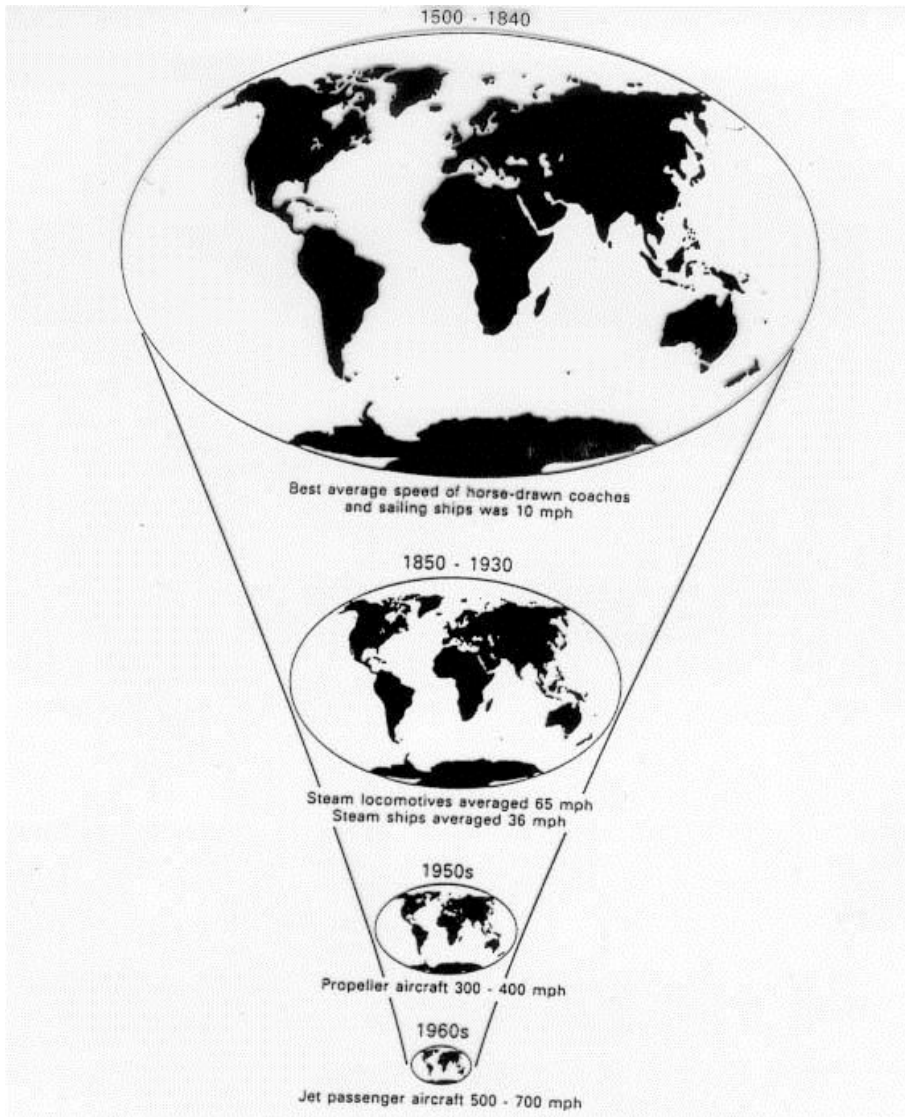
degrees of separation

=6.6 average PL

- Study on the **degrees of separation** in one of the world's largest messaging networks (2008)
- **180 million members**
- **1.3 billion chat sessions** over a one-month period
average network degree: 7.2
- Type of tie: “**communication**”

Q

What do limited degrees of separation imply for the study of innovation?



Source: Harvey (1989)

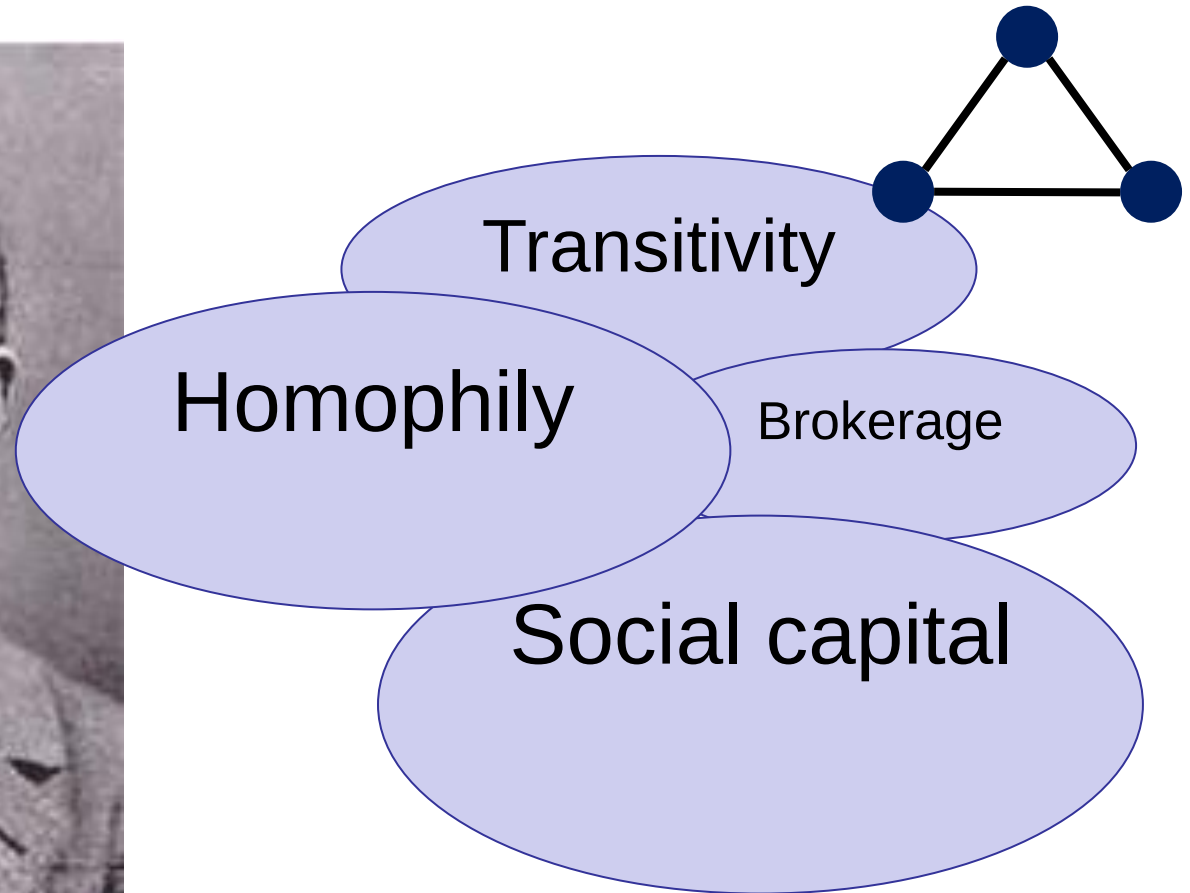
Learning objectives

At the end of this lecture, we hope you will be able...

- ...to describe the **analytical building blocks** underpinning network theorizing.
- ...understand the **assumptions** underlying **network thinking** and infer what implications this has for various types of innovation network research.
- ...interpret the **topology of networks**.

Networks in sociology

TRADITIONS OF NETWORK RESEARCH



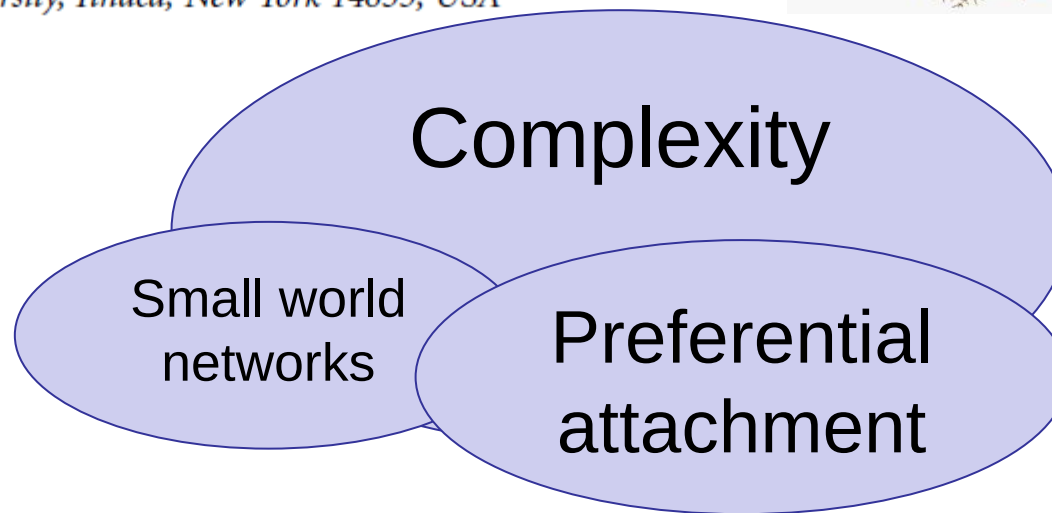
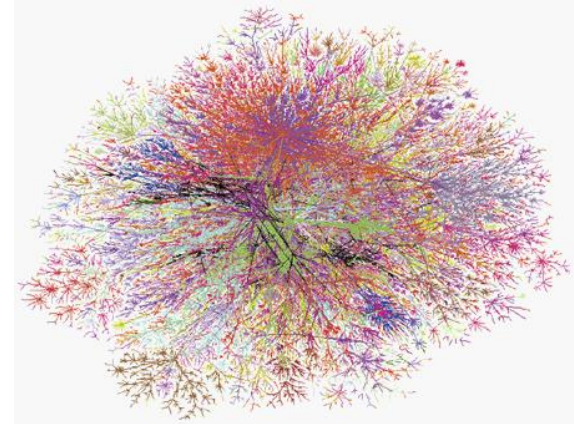
Networks in physics

TRADITIONS OF NETWORK RESEARCH

Collective dynamics of 'small-world' networks

Duncan J. Watts* & Steven H. Strogatz

*Department of Theoretical and Applied Mechanics, Kimball Hall,
Cornell University, Ithaca, New York 14853, USA*



Networks in economics & management

TRADITIONS OF NETWORK RESEARCH

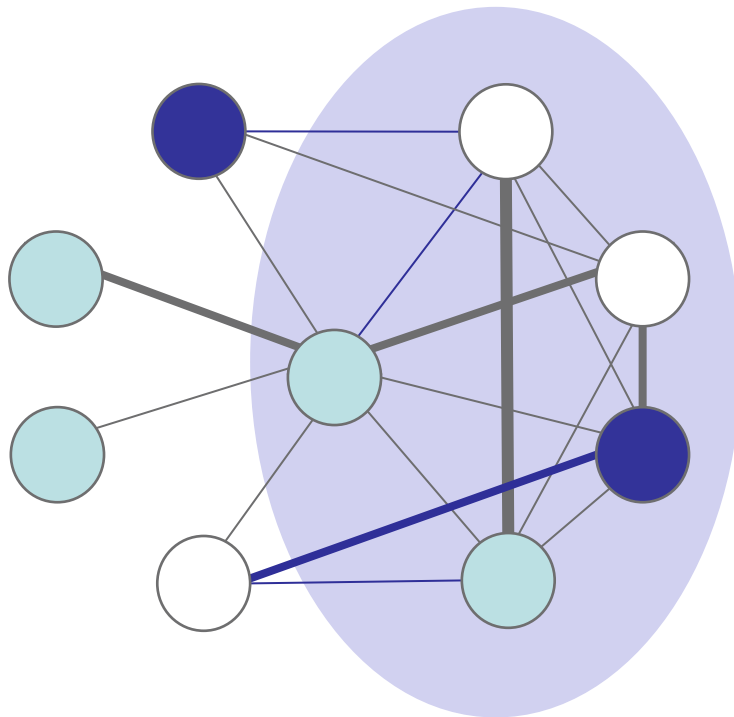
The coordination of economic action

Markets	Networks	Hierarchies
Communication through prices.	Communication through relations.	Communication through routines.
Norms established through contracts and property rights.	Norms established through complementary strengths.	Norms established through employment relations
Conflict resolution through legal action.	Conflict resolution through reciprocity and reputational concerns.	Conflict resolution through administration and supervision
High flexibility	Medium flexibility	Low flexibility
Commitment among parties is low.	Commitment among parties is medium to high.	Commitment among parties is medium to high.

Source: Williamson (1975), Powell (1990)

Network architectures

THE ANALYTICAL BUILDING BLOCKS OF NETWORK ANALYSIS

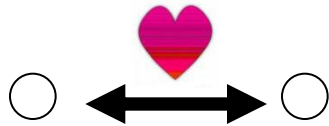


- Nodes
 - Number
 - Identity
 - Characteristics / Attributes
- Ties
 - Location
 - Content
 - Strength
- Structure
 - Pattern of ties among nodes

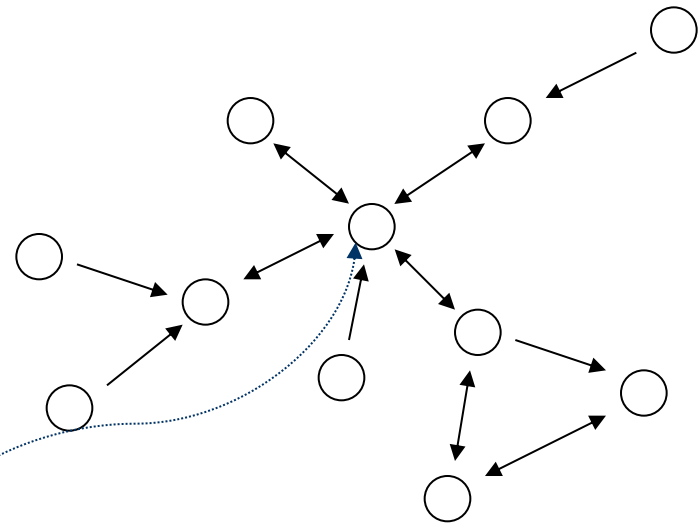
Network architectures

THE ANALYTICAL BUILDING BLOCKS OF NETWORK ANALYSIS

Relational



Structural



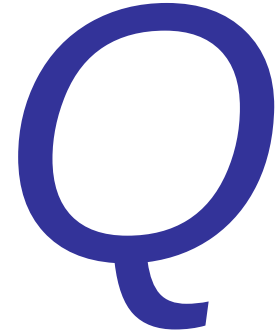
How does this network influence this person?

Dimensions of social capital

THE ANALYTICAL BUILDING BLOCKS OF NETWORK ANALYSIS

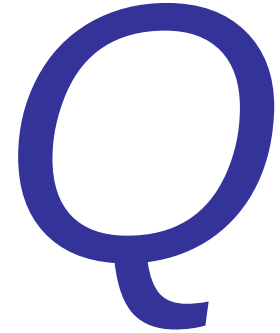
Structural	Relational	Cognitive
The structural configuration of the network	Characteristics of the relationships	Extent to which relations have shared context, language or narratives.
Influences the development of intellectual capital / information benefits	Influences the development of intellectual capital / information benefits	Influences the development of intellectual capital / information benefits
Through determining access to parties for exchanging knowledge.	Influences the conditions for exchange. - Anticipation of value - Motivation to engage	Influences the extent to which knowledge can eventually be shared.

Source: Nahapiet & Ghoshal (1998), Tsai & Ghoshal (1998)



“There is a true network out there.
It is our task as researchers to discover it”.

“Every question generates its own network”.



“Instead of capitalizing on opportunities for applying the craft, network analysts will surely advance our understanding of organization better by constructing network theories of organization”

State ties versus event ties

THE ANALYTICAL BUILDING BLOCKS OF NETWORK ANALYSIS

State ties

Ties with continuity,
but not necessarily permanent
→ *Realist perspective*

- Strength
- Duration
- Intensity

Friendship, collaboration

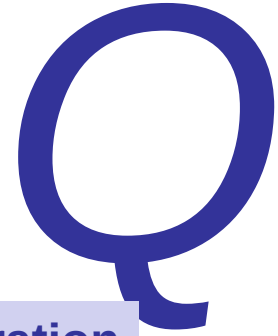
Event ties

Ties with discrete,
transitory nature
→ *Nominalist perspective*

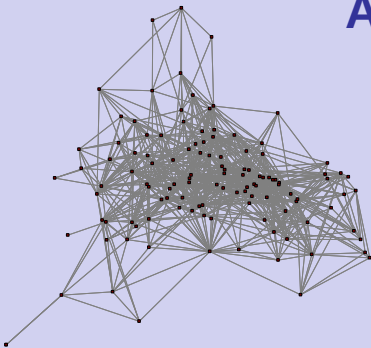
- Frequency of occurrence
- Importance

Advice seeking, email

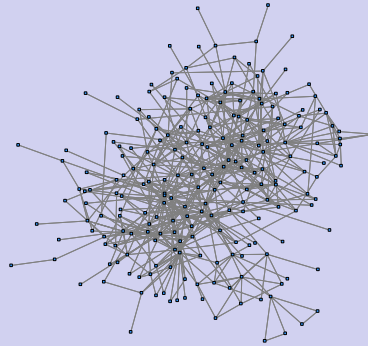
What is the meaning of **indirect paths** in the network?



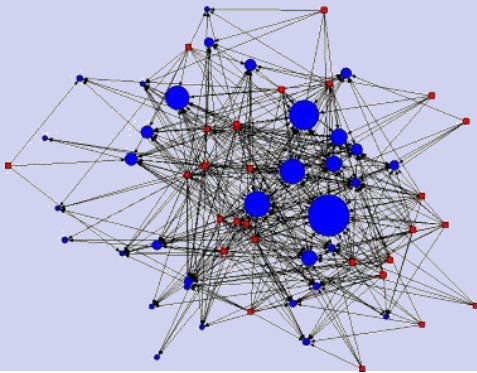
Advice-seeking



Collaboration among inventors



Labour mobility



Regional cooperation



Source lower-right image: Hoekman, Frenken , Tijssen (2010, Res Pol)

Key assumptions: the flow model

THE ANALYTICAL BUILDING BLOCKS OF NETWORK ANALYSIS

- **Key assumption:** Ties channel the flow of information
- **Structure** of the network determines how knowledge flows through the network
- **Position** in a network determines access to knowledge in the network

Labour mobility networks

THE FLOW MODEL ASSUMPTION

- **Gone but not forgotten:** enduring social relationships after labour mobility channel flows of knowledge between organizations and regions.
- **Learning-by-hiring:** organizations can learn through hiring new employees
 - Exploring technologically distant knowledge
 - Extending the hiring's firms geographic reach

Knowledge spillovers

THE FLOW MODEL ASSUMPTION

- **Investment in R&D** by private corporations and universities spills over for third-party firms to exploit.
- Spillovers have a strong **localized nature**.
- The majority of spillover studies is not concerned with the [network] **mechanisms** through which knowledge spills over.
 - Informal networks around formal collaborations may produce spillover effects
(Owen-Smith & Powell 2004)
 - Labour mobility facilitates knowledge spillovers
(Agrawal et al. 2006)
 - After controlling for labour mobility and co-invention, the residual effect of proximity on spillovers was greatly reduced
(Breschi & Lissoni 2009; Singh 2005)

Collaboration networks

THE FLOW MODEL ASSUMPTION

- **Czikzentmihályi's (1996) systems view on creativity:** creativity does not happen inside people's head, but in the interaction between a person's thoughts and the sociocultural context.
- Social ties between individuals channel the **flow** of **inspiration** and **ideas**, and **support** and **legitimacy**.
- Individuals **between the core and the periphery** of a social network are in a privileged position to achieve creative results
 - Maintain exposure to alternative sources of inspiration and ideas at the fringe
 - Without being disconnected from the base of legitimacy in the core

Networks in clusters

THE FLOW MODEL ASSUMPTION

- Firms in geographical clusters were always assumed to benefit from localized knowledge spillovers and **'local buzz'**.
- Networks in clusters may actually not be pervasive and collective, but **uneven** and **selective**.
- Inequality in network position drive differences in access to knowledge and are therefore reflected in **performance differences** between firms.
- **Global pipelines** (non-local collaboration ties) shown to have an additional and sometimes stronger effect on innovation performance.

Learning objectives

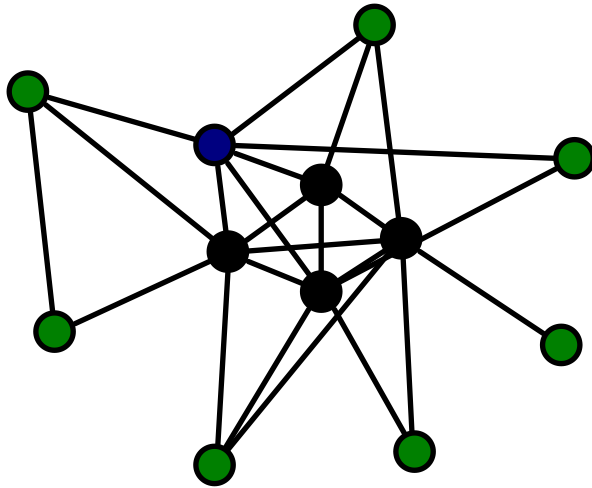
At the end of this lecture, we hope you will be able...

- ...to describe the **analytical building blocks** underpinning network theorizing. 
- ...understand the **assumptions** underlying **network thinking** and infer what implications this has for various types of innovation network research. 
- ...interpret the **topology of networks** and appreciate its role in the **comparison** of networks.

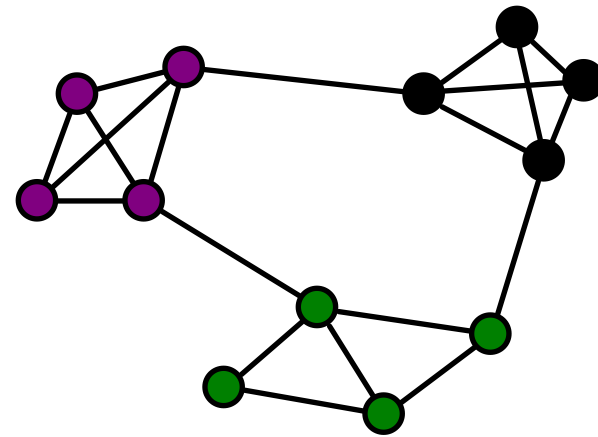
Average path length

THE TOPOLOGY OF NETWORKS

Average geodesic distance = 1.9
“core-periphery”



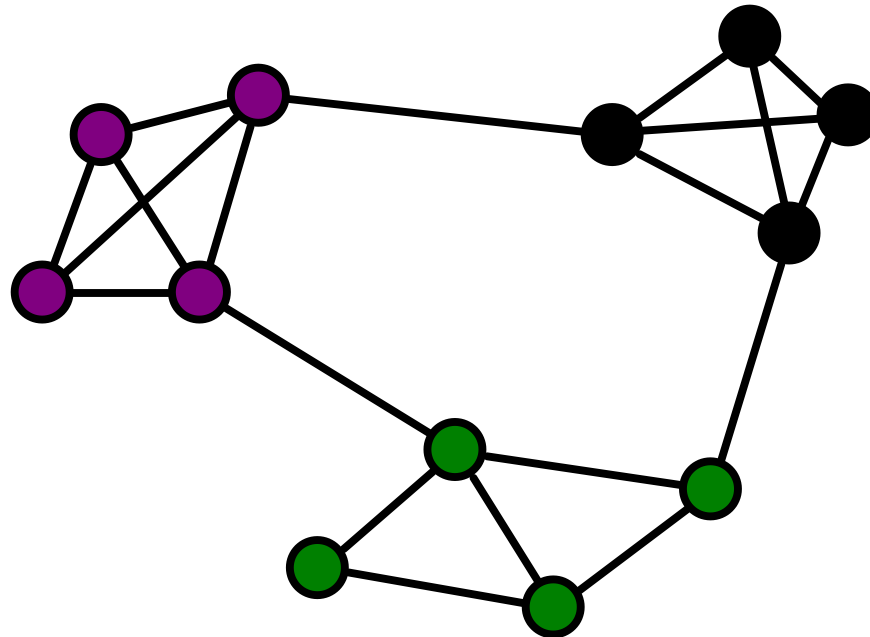
Average geodesic distance = 2.4
“clique structure”



Average path length = Characteristics path length = Average geodesic distance

Clustering coefficient

THE TOPOLOGY OF NETWORKS



Average clustering coefficient = 0.777

The example of Sophia-Antipolis

THE TOPOLOGY OF NETWORKS



Artificially created 'business park', established in the late 1970s: Europe's Silicon Valley

Mostly IT firms with activities ranging infrastructure, platforms and applications

Many pharmaceutical firms with activities ranging from drugs, cosmetics, and fine chemistry

Many public and private research institutes

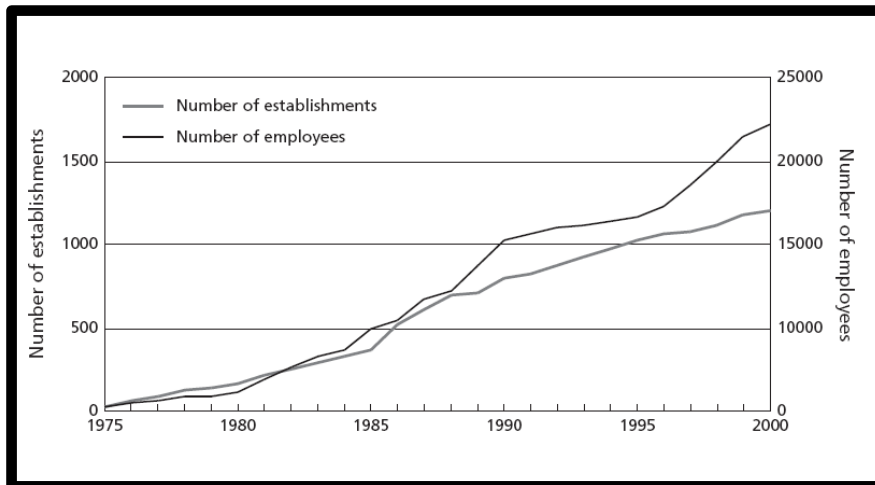


Source: Ter Wal (2013)



The example of Sophia-Antipolis

THE TOPOLOGY OF NETWORKS



	Information Technology	Pharma
Emergence	High growth Foreign multinationals	High growth Foreign multinationals
Growth	High growth Spin-offs / High-tech start-up	Low growth Foreign multinationals

Source: Ter Wal (2013)

**To what extent did
a network of collective learning
emerge in Sophia-Antipolis
Among scientists and engineers
in Life Sciences and Information Technology?**

Source: Ter Wal (2013)

The example of Sophia-Antipolis

THE TOPOLOGY OF NETWORKS



US 20080068132A1

(19) **United States**

(12) **Patent Application Publication** (10) **Pub. No.: US 2008/0068132 A1**
Kayanakis et al. (43) **Pub. Date: Mar. 20, 2008**

(54) **CONTACTLESS RADIOFREQUENCY
 DEVICE FEATURING SEVERAL ANTENNAS
 AND RELATED ANTENNA SELECTION
 CIRCUIT**

(76) Inventors: **Georges Kayanakis**, Antibes (FR);
Yannick Grasset, Vallauris (FR); **Elias
 Sabbah**, Valbonne (FR)

Correspondence Address:
JAMES C. LYDON
100 DAINGERFIELD ROAD
SUITE 100
ALEXANDRIA, VA 22314 (US)

(21) Appl. No.: **11/798,456**

(22) Filed: **May 14, 2007**

(30) **Foreign Application Priority Data**

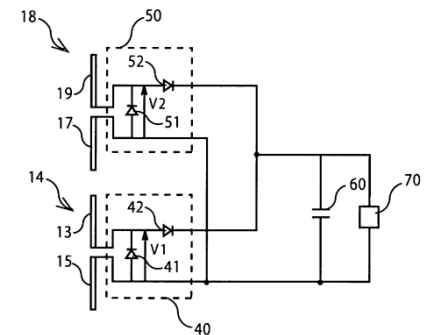
May 16, 2006 (FR)..... 0604363
 May 16, 2006 (FR)..... 0604365

Publication Classification

(51) **Int. Cl.**
H04Q 5/22 (2006.01)
 (52) **U.S. Cl.** **340/10.1**

(57) **ABSTRACT**

The invention relates to an integrated circuit for contactless radiofrequency device connected to a first antenna and to a second antenna designed to receive a radiofrequency signal coming from a reader. According to a main characteristic, the integrated circuit includes a first rectifier circuit and a second rectifier circuit to rectify each radiofrequency signal received from the first antenna and the second antenna, respectively, so as to produce two positive output voltages **V1** and **V2**, the rectifier circuits being mounted in parallel in order to select an output voltage value that corresponds to the maximum voltage value between **V1** and **V2**.



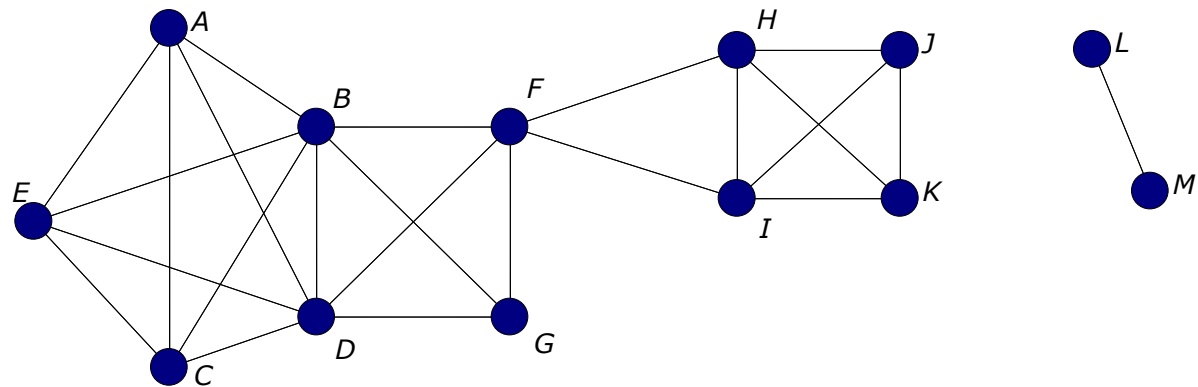
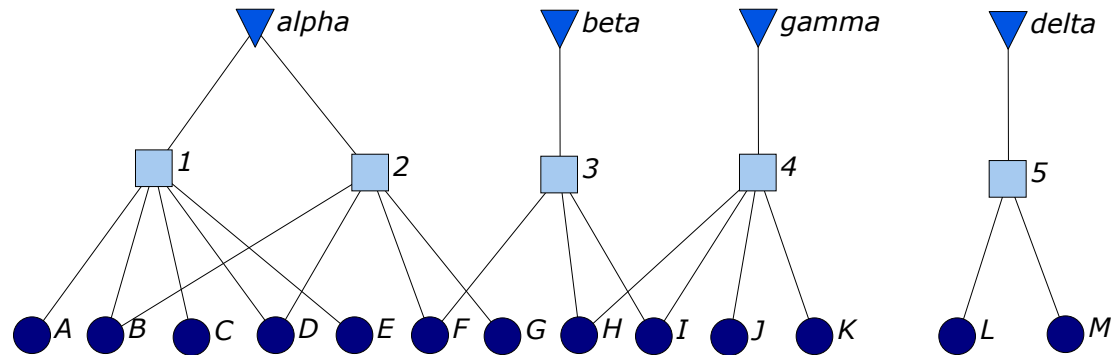
The example of Sophia-Antipolis

THE TOPOLOGY OF NETWORKS

Applicants

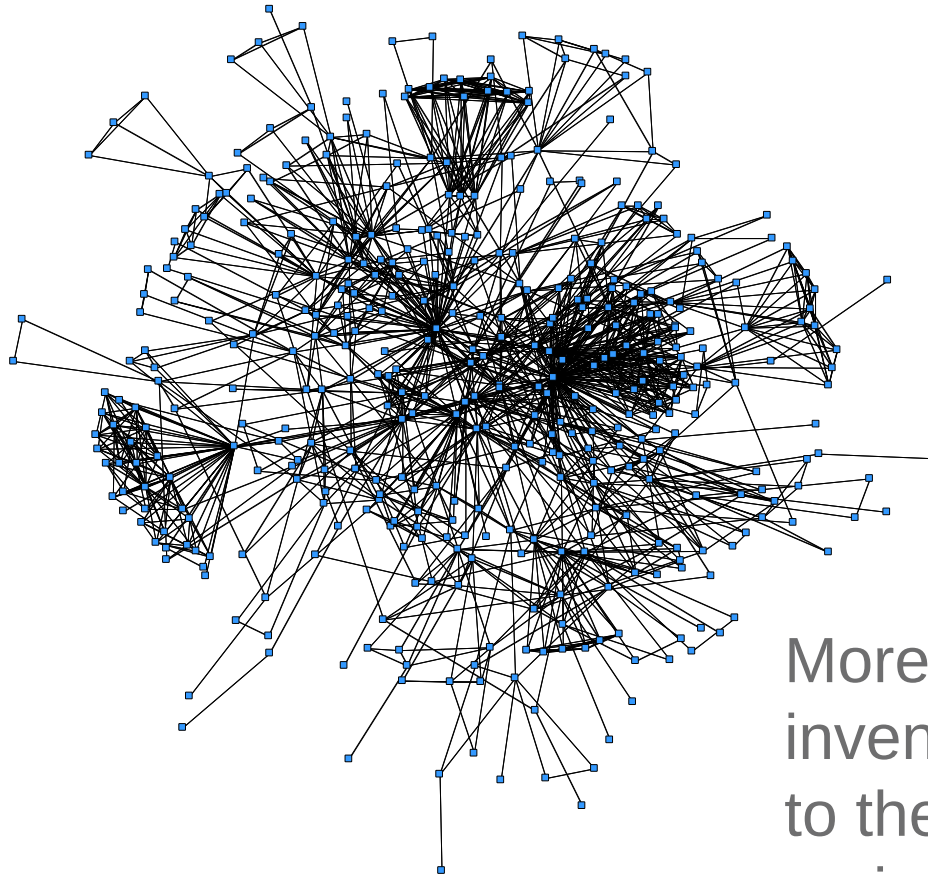
Patents

Inventors



The network in ICT

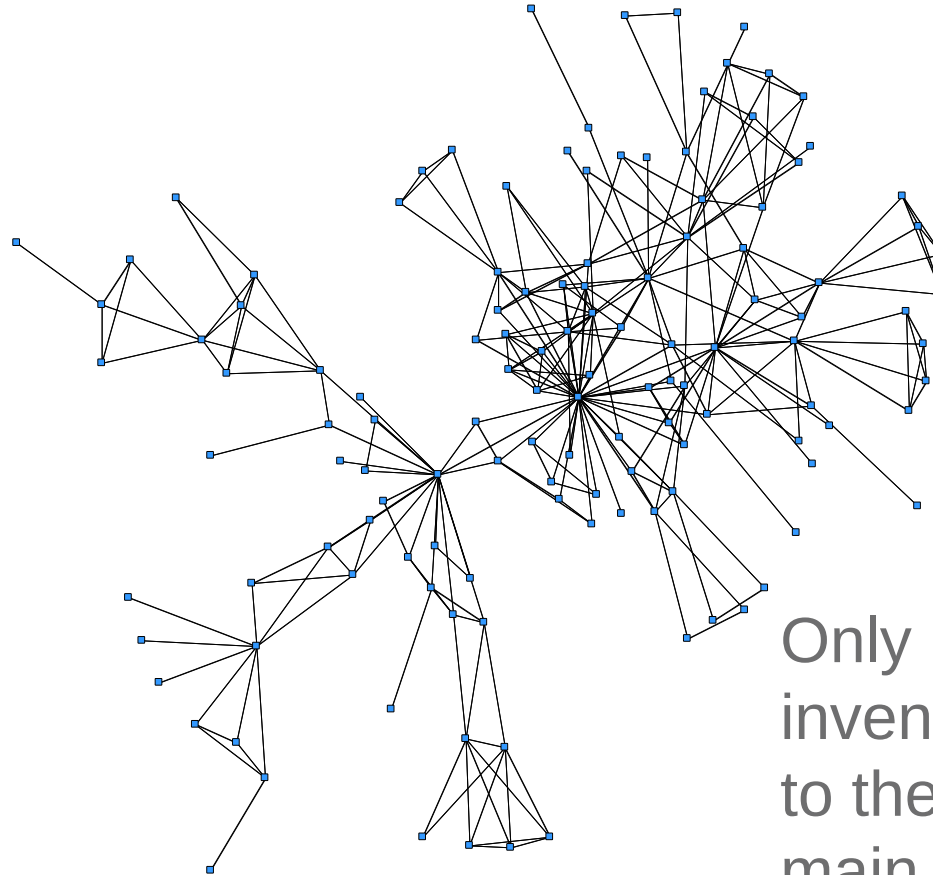
ICT



More than 30% of inventors belong to the networks main component

The network in pharma

Pharma

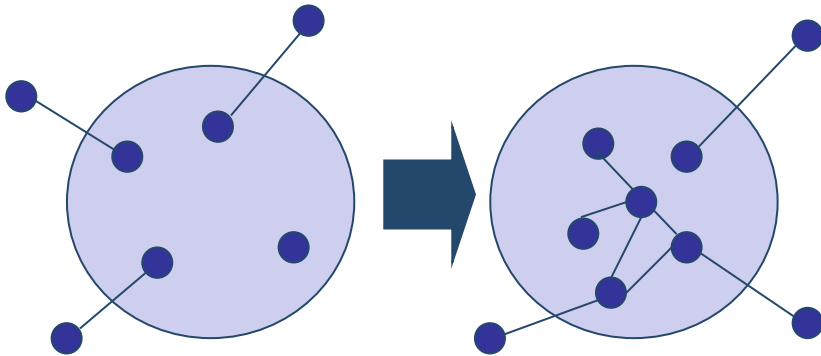


Only 10% of
inventors belongs
to the networks
main component

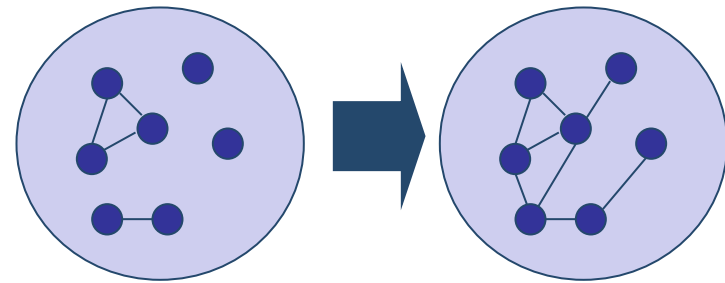
What are the implications of network growth and two-mode structure for measuring these factors?

Q

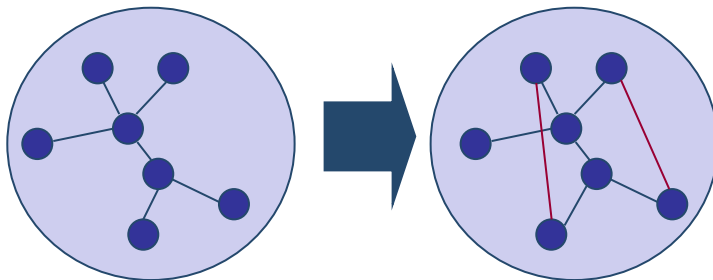
Geographical orientation



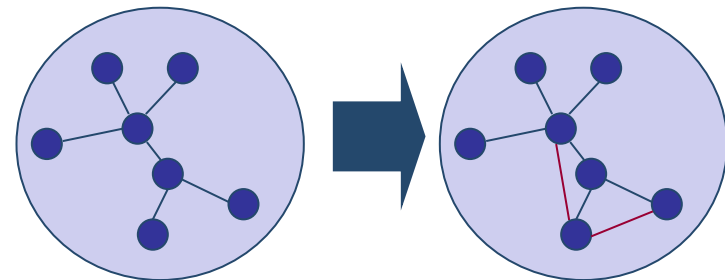
Connectivity



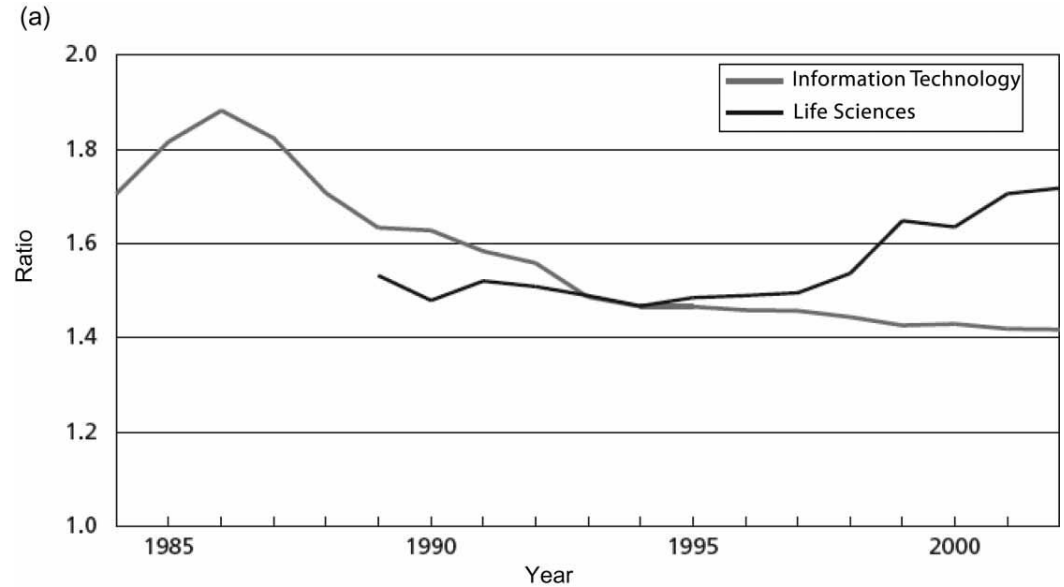
Average path length



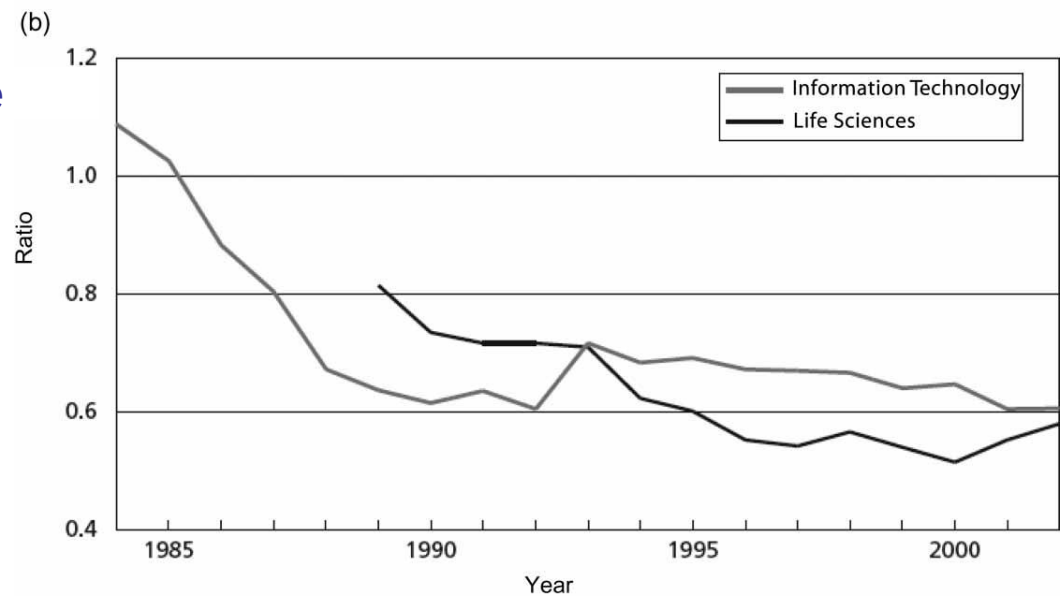
Clustering coefficient



Path length ratio over time



Clustering coefficient ratio over time





Over the course of time
a network of inventors and engineers
who collaborate across firm boundaries
only emerged in Information Technology
and not in the Pharmaceutical Sector

Network size

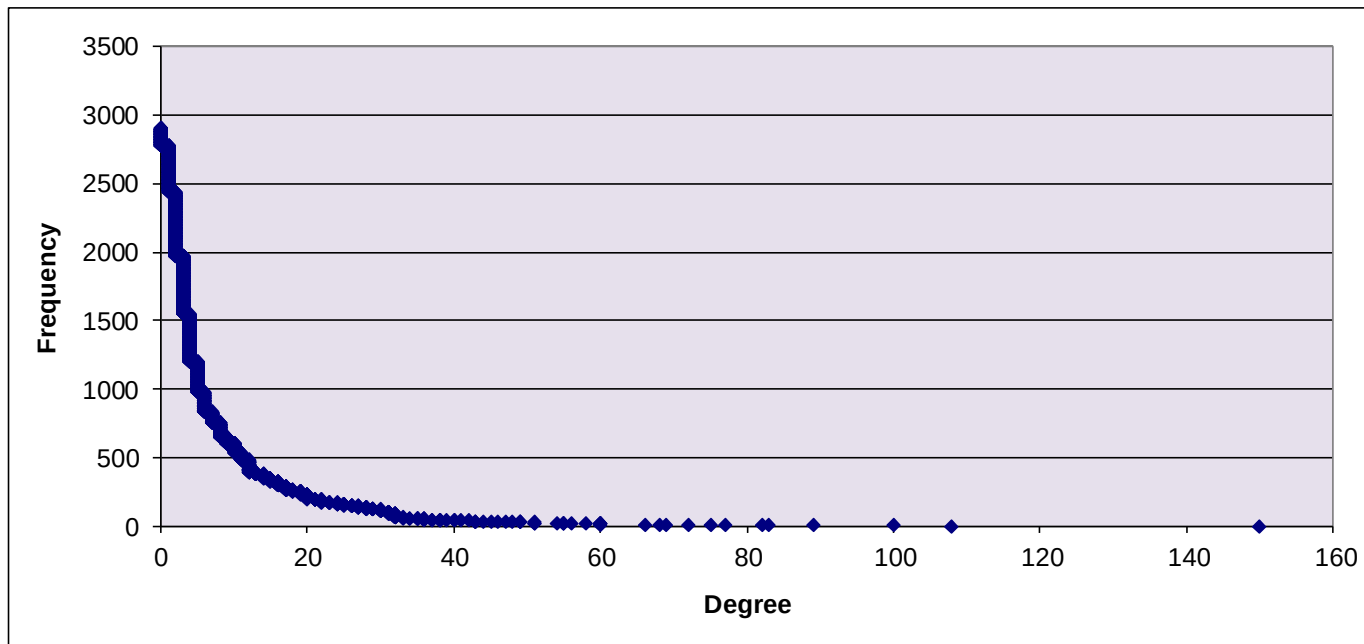
THE TOPOLOGY OF NETWORKS

Comparing networks is a tricky business.

- Density tends to decrease if network size increases
 - The number of ties increases linearly
 - The number of dyads increases quadratically
- Most network properties are highly sensitive to network size.
- Two-mode structure underlying many networks influences network topology.

Degree distribution

THE TOPOLOGY OF NETWORKS



$\langle k \rangle$

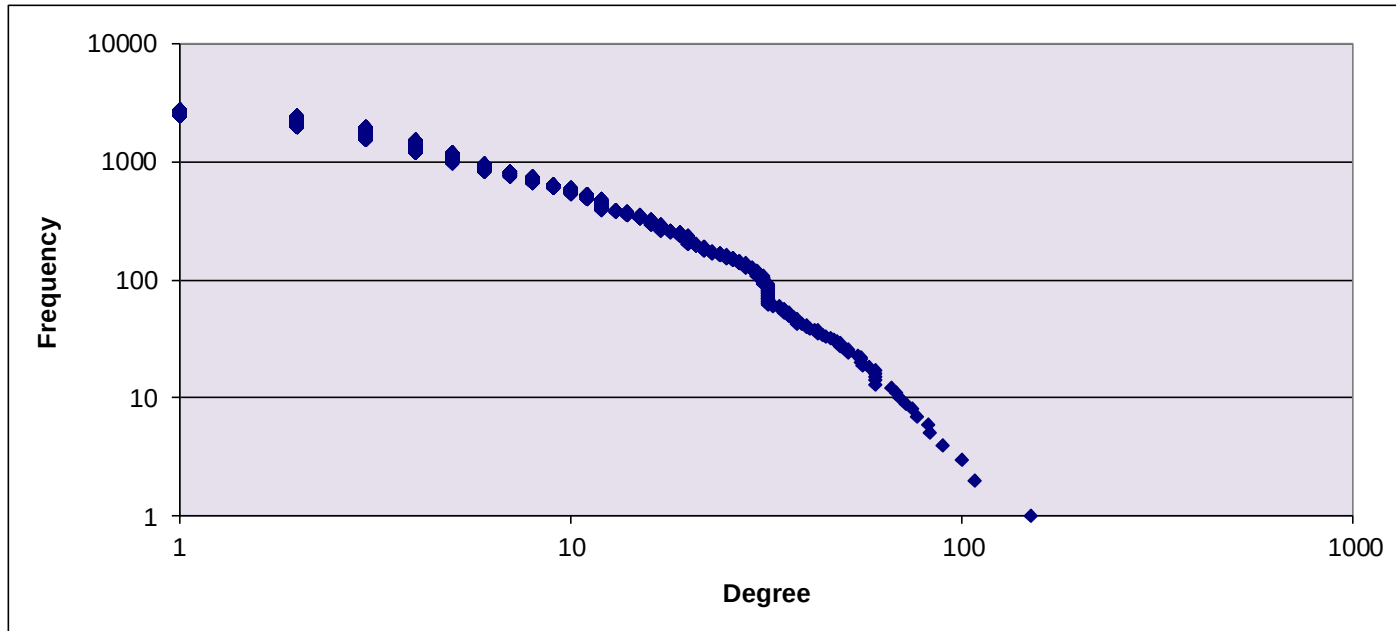
The average degree

$P(k)$

The degree distribution of a graph indicating the probability that a node randomly extracted has exactly k connections

Degree distribution

THE TOPOLOGY OF NETWORKS



$\langle k \rangle$

The average degree

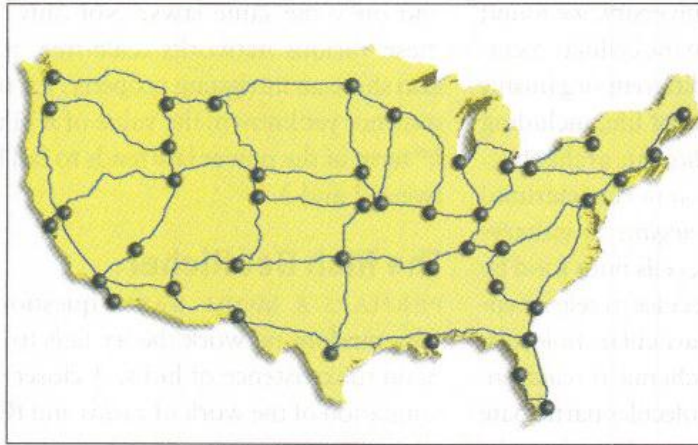
$P(k)$

The degree distribution of a graph indicating the probability that a node randomly extracted has exactly k connections

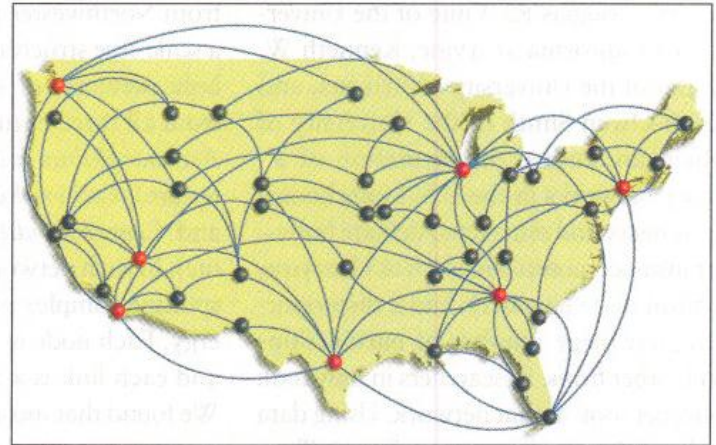
Scale-free networks

Degree distribution

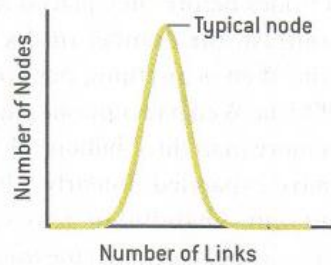
Random Network



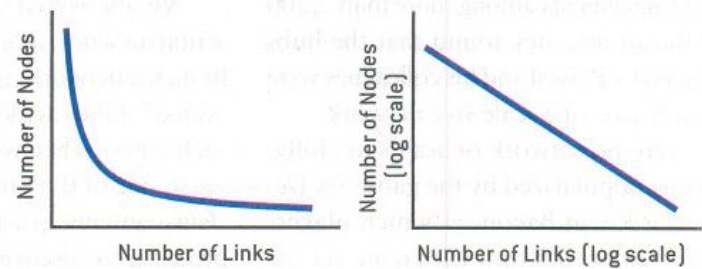
Scale-Free Network



Bell Curve Distribution of Node Linkages



Power Law Distribution of Node Linkages



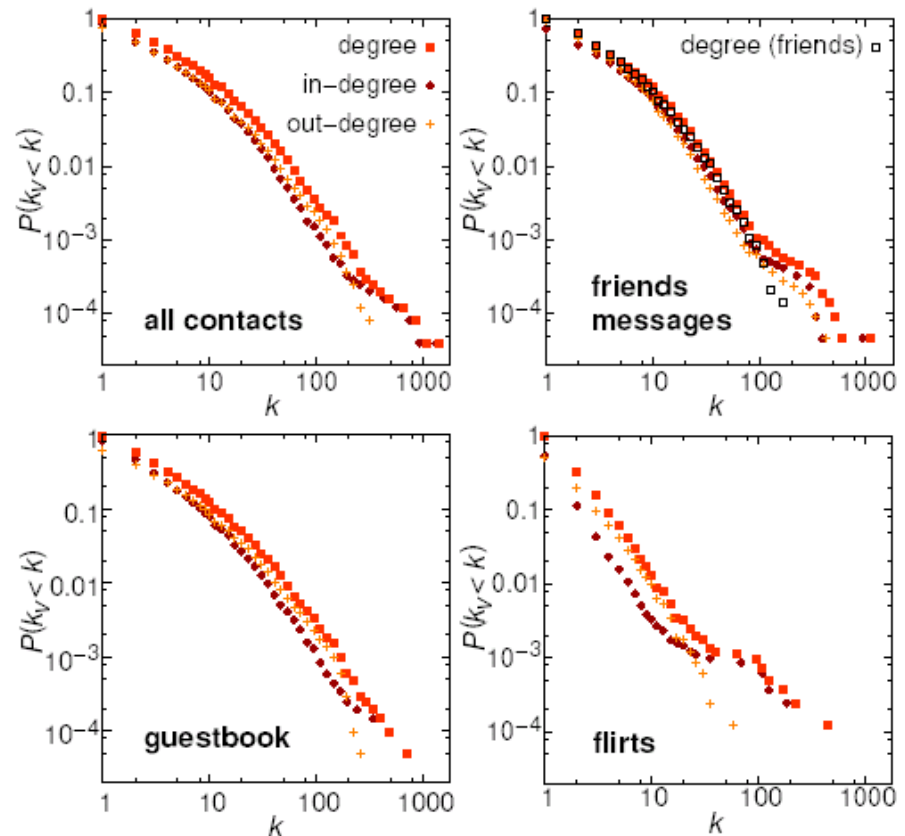
Scale-free networks

Degree distribution

Online community

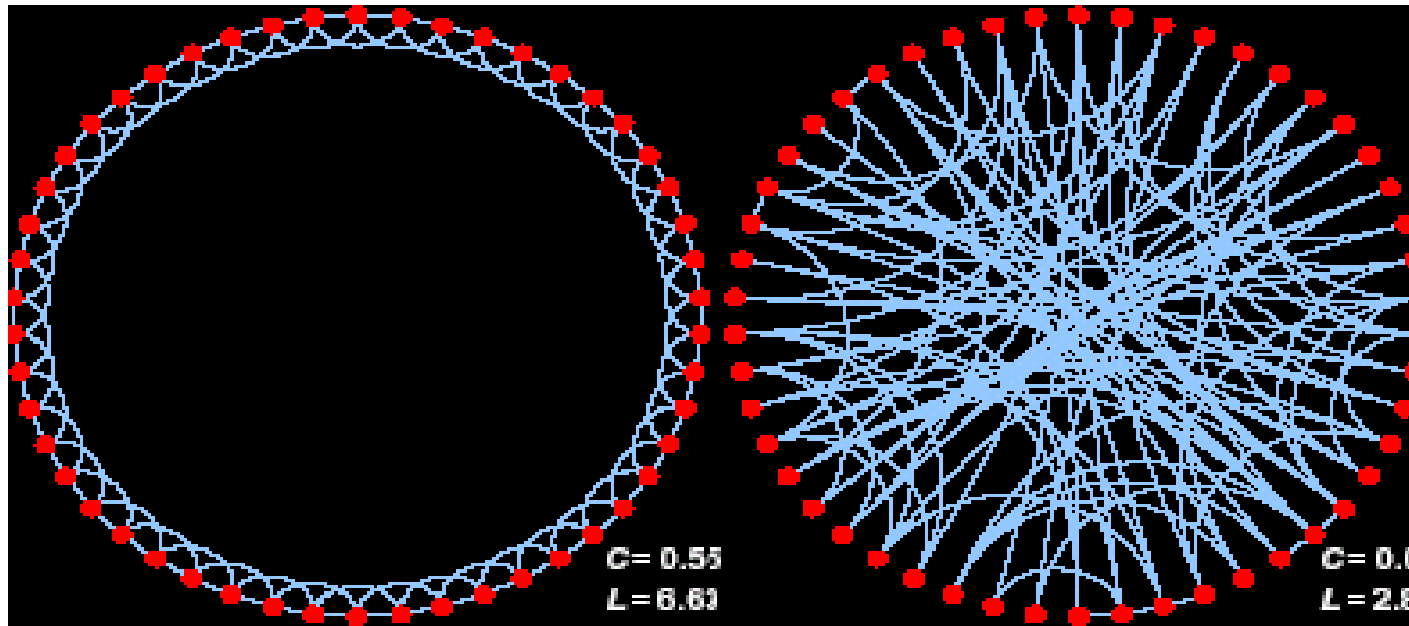
512 days

25.000 users



Regular and random graphs

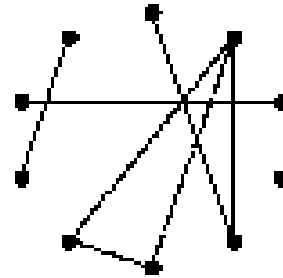
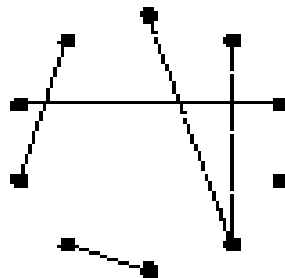
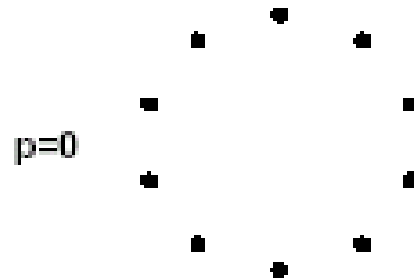
THE TOPOLOGY OF NETWORKS



Two extreme networks

Random graphs

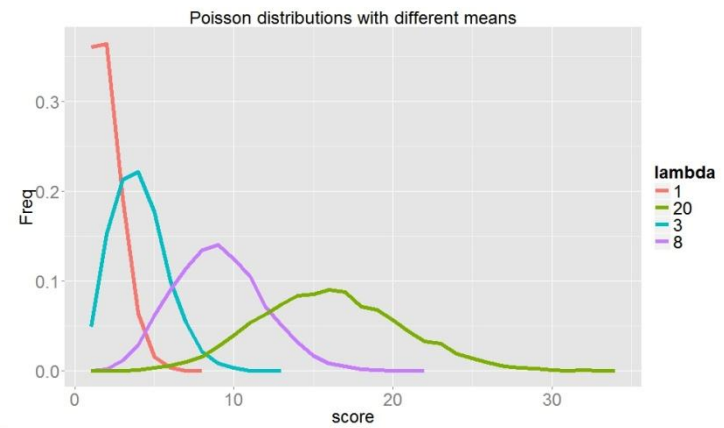
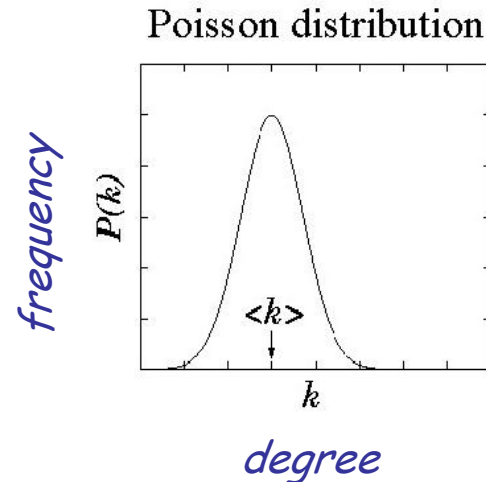
THE TOPOLOGY OF NETWORKS



Random graphs

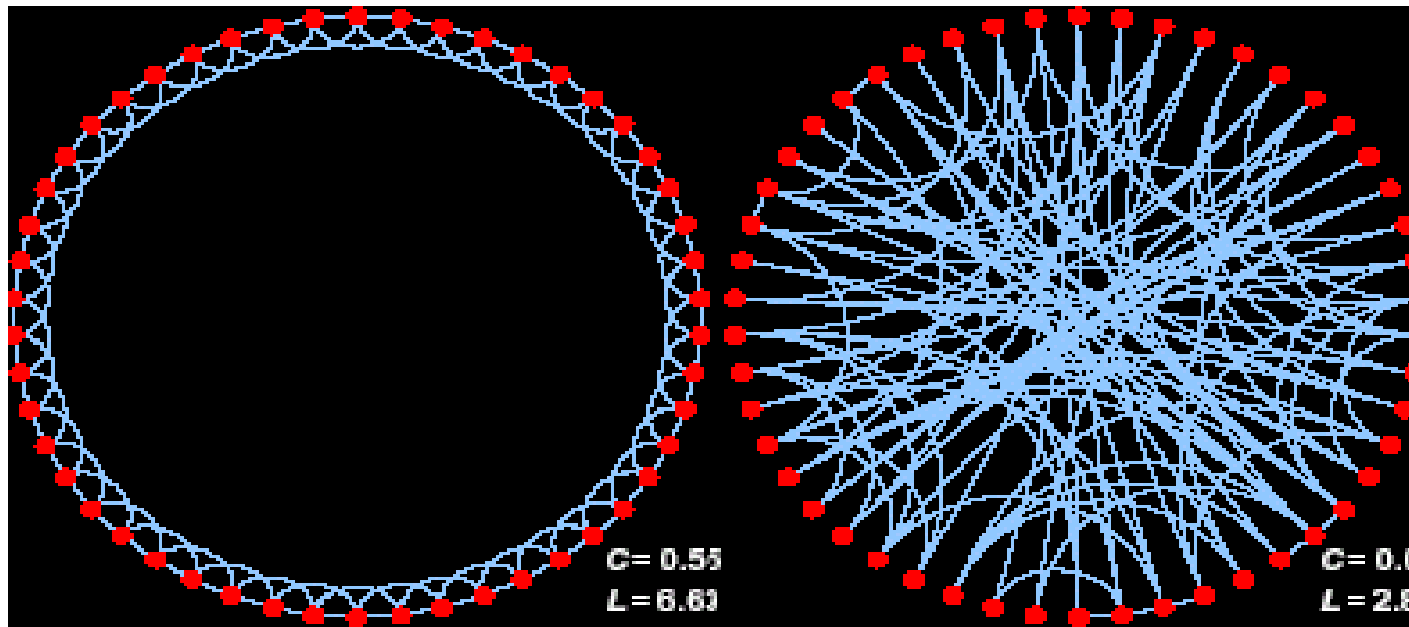
THE TOPOLOGY OF NETWORKS

Erdős and Rényi demonstrated that the degree distribution for a random graph follows a Poisson's distribution



Regular and random graphs

THE TOPOLOGY OF NETWORKS



PL *very high*

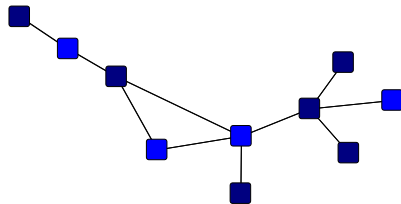
CC *very high*

PL *very low*

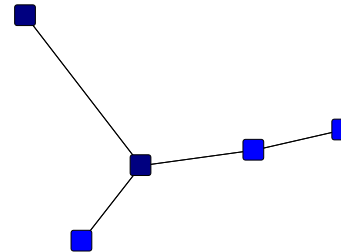
CC *very low*

Random networks

Of equal size and with equal degree distribution
are the basis of comparison in
social network analysis



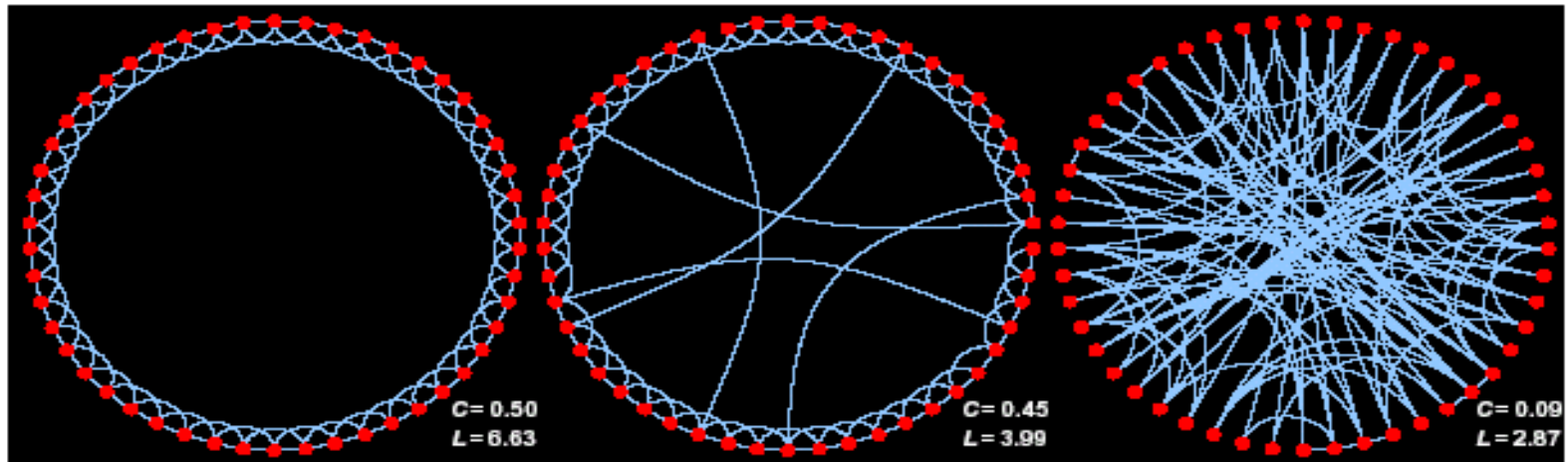
Average PL: 2.489
PL in random network of equal size: 3.321
PL ratio (actual / random PL) 0.749



Average PL: 1.800
PL in random network of equal size: 1.641
PL ratio (actual / random PL) 1.097

Small world networks

THE TOPOLOGY OF NETWORKS



$$p=0$$

Regular Network

High PL

High CC

$$0 < p < 1$$

Small-World Network

Low PL

High CC

$$p=1$$

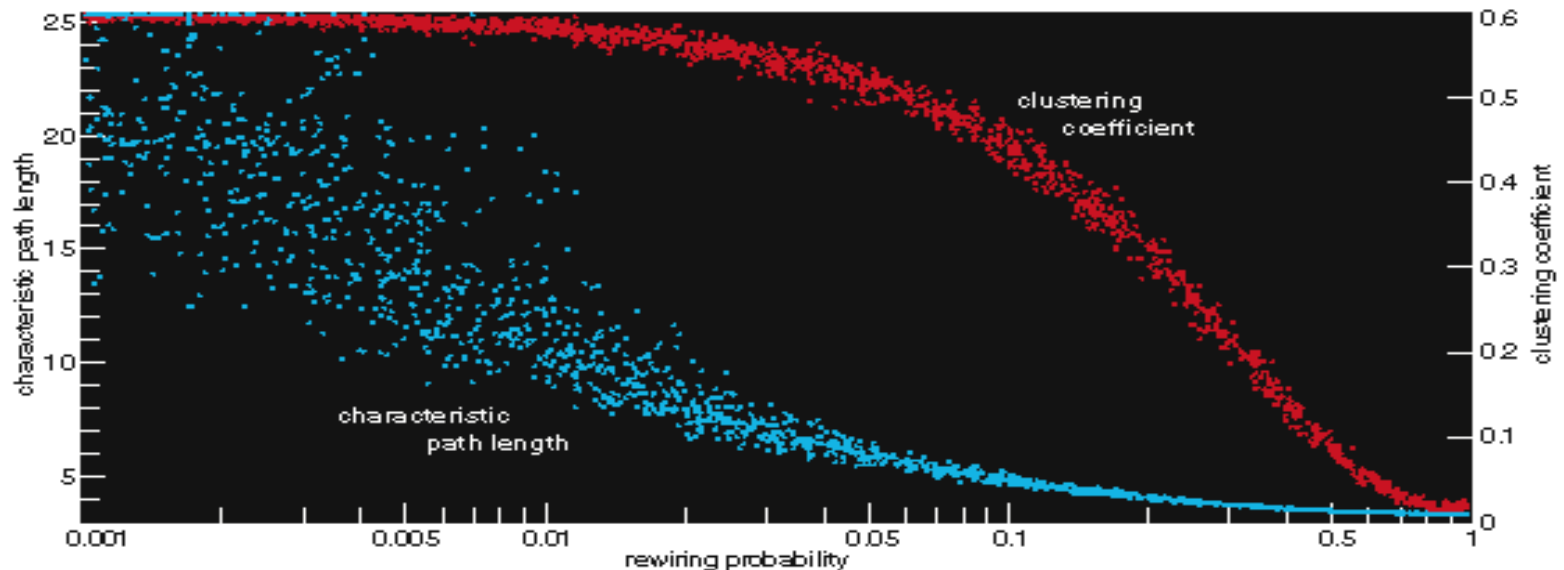
Random Network

Low PL

Low CC

Small world networks

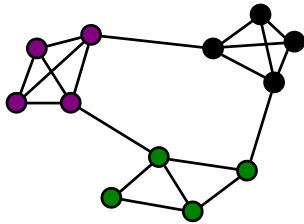
THE TOPOLOGY OF NETWORKS



Rewiring probability

Detecting small world networks

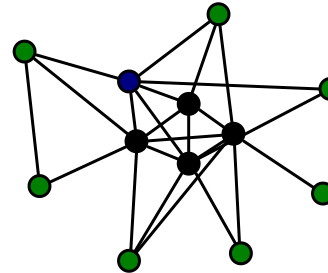
THE TOPOLOGY OF NETWORKS



$$PL_{actual} = 1.9$$

$$PL_{random} = 2.1$$

$$PL_{ratio} = 0.9$$



$$PL_{actual} = 2.4$$

$$PL_{random} = 1.8$$

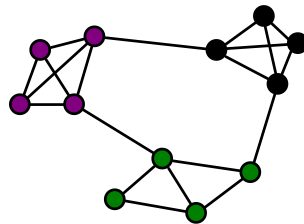
$$PL_{ratio} = 1.3$$

Small world:

- If the PL in the real network is as short in a random network of equal size
- In other words: if the PL-ratio is close to 1

Detecting small world networks

THE TOPOLOGY OF NETWORKS



$$CC_{actual} = 0.777$$

$$CC_{random} = 0.303$$

$$CC_{ratio} = 2.564$$

Small world:

- If the CC in the real network is much higher than in a random network of equal size
- In other words: if the CC-ratio is far from 1

Small world networks

THE TOPOLOGY OF NETWORKS

	L	L_{rand}	C	C_{rand}	N
WWW	3.10	3.35	0.11	0.00023	153127
Film actors	3.65	2.99	0.79	0.00027	225226
Power Grid	18.7	12.4	0.08	0.005	4014
C. Elegans	2.65	2.25	0.28	0.05	282

Learning objectives

At the end of this lecture, we hope you will be able...

- ...to describe the **analytical building blocks** underpinning network theorizing. 
- ...understand the **assumptions** underlying **network thinking** and infer what implications this has for various types of innovation network research. 
- ...interpret the **topology of networks** and appreciate its role in the **comparison** of networks. 

Implications for your research

Some **advice** for your own research:

- Comparing networks is **not a purpose on itself**.
- When comparing properties across networks or over time you will need to take the topology of your network into account, most often by comparing against a **baseline random network**.
- You will need to understand the **assumptions** that underpin network theories and concepts and the implications this has for the networks you study.

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